

Air University



Object Oriented Programming (Lab)

Year 2025

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Lab Task 05

Program

Code

```
1  ∨ #include <iostream>
2    #include <cstring>
3    using namespace std;
4
5  ∨ class student
6    {
7    private:
8        string *std_name;
9        int std_age;
10
11    public:
12  ∨    student()
13        {
14            *std_name = "Unknown";
15            std_age = 0;
16        }
17  ∨    student(string name, int age)
18        {
19            std_name = new string(name);
20            std_age = age;
21        }
22  ∨    student(student &obj)
23        {
24            std_name = new string (*(obj.std_name));
25            std_age = obj.std_age;
26        }
27    void display()
28    {
29        cout << "Student Details: " << endl;
30        cout << "Name: " << *std_name << endl;
31        cout << "Age: " << std_age << endl;
32    }
33    ~student()
34    {
35        delete std_name;
36        cout << "Destructor is called. " << endl;
37    }
38 };
39
```

```
40  int main()
41  {
42      student s1("Usama", 20);
43      s1.display();
44
45      student s2(s1);
46      s2.display();
47
48      cout << endl;
49      return 0;
50  }
```

Output

```
Student Details:
Name: Usama
Age: 20
Student Details:
Name: Usama
Age: 20

Destructor is called.
Destructor is called.
```

THE END
